Agentic Cloud Cluster

Sarthak Siddhpura - AU2240041

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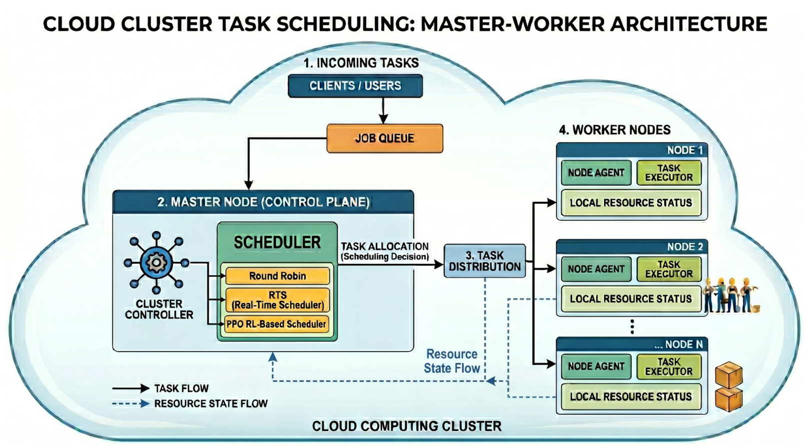
Internal mentor: Professor Sanjay Chaudhary

Ahmedabad University

Introduction

Agentic Cloud Cluster is a distributed task scheduling system that intelligently allocates containerized workloads across compute clusters. It demonstrates how adaptive scheduling strategies from traditional approaches to reinforcement learning-driven methods can significantly improve task execution efficiency compared to standard methods



Results

Analysis

reward func optimizing SLA compliance, queue latency, and cluster load balancing via production trace data

$$R(s,a) = \underbrace{1.4}_{\text{base}} + \underbrace{0.25\,H}_{\text{headroom}} - \underbrace{0.35\,Q_p}_{\text{queue}} - \underbrace{0.55\,T_p}_{\text{tail} \star} - \underbrace{0.20\,I_p}_{\text{hotspot}} - \underbrace{0.40\,\Delta I}_{\text{balance}} - \underbrace{R_q}_{\text{requeue}}$$

- $Q_p = \min((q_w + t_r \bar{L})/k t_r, 3)$
  Queue pressure relative to SLA budget $k \cdot t_r$.
- $I_p = \max(\bar{L}(w) - \mu_{cluster}, 0)$
  Hot-spot penalty; discourages overloading a single worker.
- $R_q = \min(n_{requeue}, 4) \times 0.05$
  Requeue penalty for rescheduled tasks.
- $H = 1 - \bar{L}(w)$
  Headroom bonus. $\bar{L}(w) = \min(⅓(C+M+S \text{ utilization}), 1.5)$ is normalized load.
- $T_p = \max((q_w + 2 t_r \bar{L})/k t_r - 1, 0)$
  Tail-latency risk penalty (highest weight: 0.55).
- $\Delta I = \sigma_{after} - \sigma_{before}$
  Load-balance improvement; rewards reducing std-dev of loads.

Infeasible placement $\rightarrow R = -1.8$ (hard constraint violation)

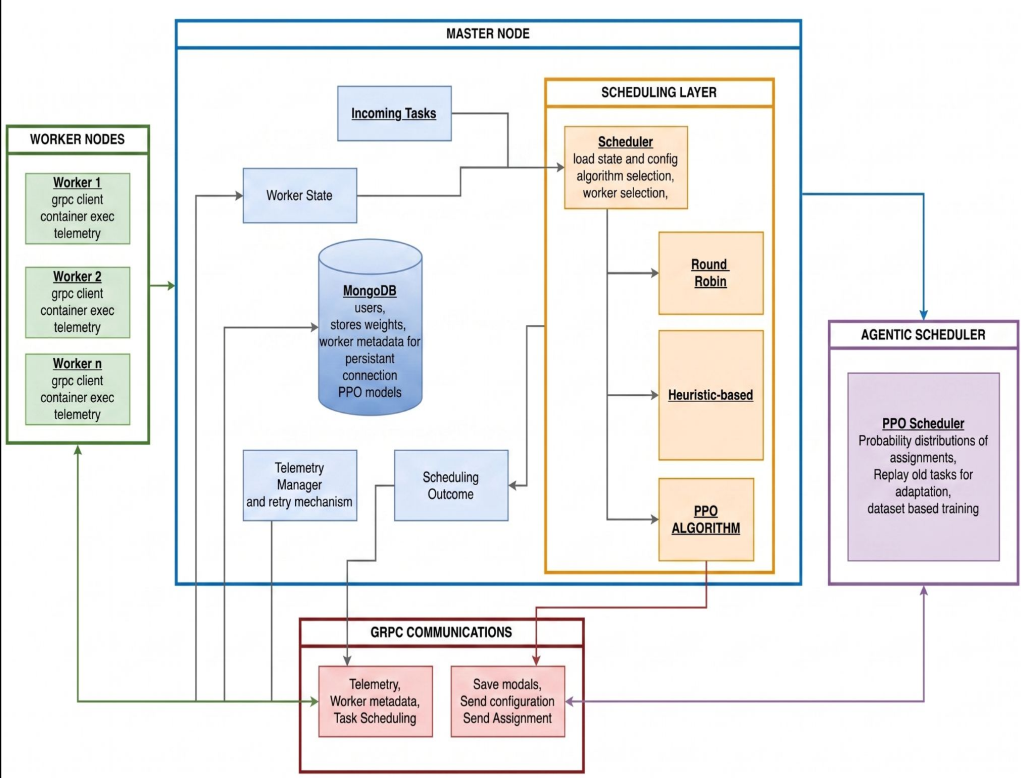
Conclusions

Reinforcement learning driven scheduling (PPO) adapts more effectively to varying workload patterns and achieves measurable improvements in task latency compared to traditional baselines. This demonstrates that reinforcement learning-based schedulers are viable for real distributed systems and can continuously adapt to changing cluster conditions.

Contributions

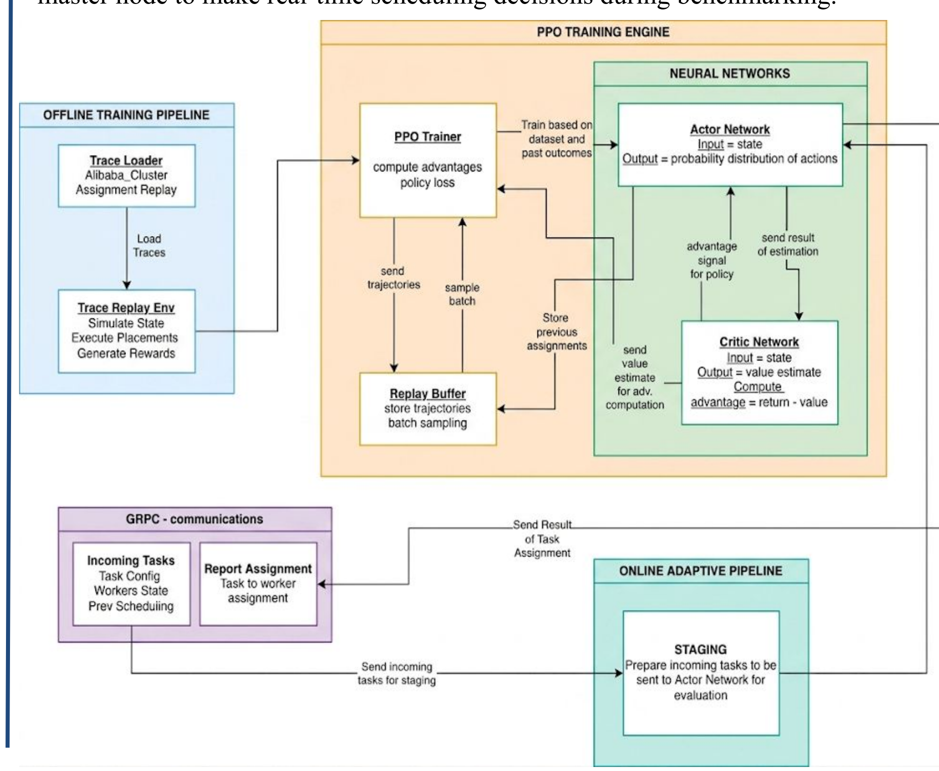
developed a distributed task scheduling system with multiple scheduling algorithms and created a comprehensive benchmarking framework that demonstrates reinforcement learning-driven scheduling achieves superior performance compared to traditional approaches.

Methodology



Master-worker distributed system where the master makes scheduling decisions and workers execute containerized tasks, communicating via gRPC.

The **PPO scheduler** is trained offline using the agentic_scheduler module, which replays historical task traces through a reinforcement learning environment to learn optimal worker assignment policies. This trained model is then deployed in the master node to make real-time scheduling decisions during benchmarking.



References

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